

RAMCO INSTITUTE OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING
ACADEMIC YEAR 2019-20
ME6811-PROJECT WORK

Sl. No.	Project title	PO's relevance	PSO's relevance	Outcome
New Product Development				
1.	Experimental Study on Spray Flash Distillation	1,2,3,4,5,6,7,8,9,10,11,12	1,2,3	Best project in Science Day 2020 Competition (I prize), Conference presentation (M.Rajagopal, A.Santhan Kumar, M.Saravana Kumar, S.Srijith, R.Arun Kumar, Experimental Study on Spray Flash Distillation, International Web Conference on Smart Engineering Technologies, Ramco Institute of Technology, 26, 27, June, 2020)
2.	Experimental Study of effective Throttling in Closed Loop Heat Pump Dryer	1,2,3,4,5,6,7,8,9,10,11,12	1,2,3	Best project in Science Day 2020 Competition (II prize)
3.	Designing a Fixture for Guiding the Front Shell of Vacuum Booster Assembly in Actuation Unit	1,2,3,4,5, 6, 8,9,10,11,12	1,2	The work has been completed and converted as a paper. Yet to be published
4.	Conversion of Rubber Waste into Commercial Automotive Components	1,2,3,5,6,8,9,10,11,12	2	S Ramar M Sankaran presented paper in International Web Conference on Smart Engineering Technologies 2020 (IWCSET -2020), Ramco Institute of Technology, Rajapalayam.
5.	Mechanized Drainage Cumseaside Cleaning System By using Solar System	1,2,3,4,5,9,10,11,12	1,4	The work has been suggested to apply for a patent with some improvement.
Projects in Identified Researches				
6.	Characterization and Machining Studies on Hybrid Aluminium Composites In AWJM	1,2,3,5,6,8,9,10,11,12	2	The work will be converted as a conference or Journal paper in the forthcoming academic year.

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7.	Imposing severe plastic deformation on CP-Ti for producing ultrafine grain through Repetitive corrugation and straightening technique for biomedical applications	1,2,3,5,6,8,9,10,11,12	2	Dr.S.Rajakarunakaran, Mr.P.Sureshkumar, Mr.A.Dhineshraj,Mr.P.Muthukumar,Mr.R.Muthukumar, "Imposing severe plastic deformation on CP-Ti for producing ultrafine grain through repetitive corrugation and straightening technique for Biomedical applications", International Web conference on Science, Engineering & Technology (IWCSET-2020), May 15-16, 2020
8.	Finite Element Modelling And Experimentation On CFRP And Glare Sandwich Composites For Warship Hull Under Static And Dynamic Loading	1,2,3,4,5,6,7,8,9,10,12	1,2	1. J.Jerold John Britto, R.Venkatesh, K.Amudhan, A.Vasanthanathan, N.Gokulakrishnan, M.Manikandan, “Micromechanics study on FRP composite cylinder under finite element simulation COMSOL Multiphysics®”, Materials Today Proceedings, 1 May 2020, https://doi.org/10.1016/j.matpr.2020.02.813, 2. J. Jerold John Britto, N. Gokulakrishnan ,M. Manikandan, Dhanajayan, ” Finite Element Modeling And Simulation Of Cfrp /Glare Composite For Warship Hull Under Static And Dynamic Condition” ISTE Sponsored 6th International E-Conference on Latest Trends in Science, Engineering and Technology, (ICLTSET’20) on April 13, 2020 Organized by Karpagam Institute of Technology, Coimbatore. 3. Jerold John Britto, J., R. Venkatesh, K. Amudhan, A. Vasanthanathan, N. Gokulakrishnan, and M. Manikandan, “Micromechanics study on FRP composite cylinder under finite element simulation COMSOL Multiphysics®” International Conference on Advances in Materials Research – 2019 at Bannari Amman Institute of Technology, Sathyamangalam, TN on December 06&07, 2019.

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9.	Analysis of surface properties of Plasma Sprayed composite coating (50% WC + 50% 8YSZ) on Austenitic Stainless Steel (SS316)	1,2,3,4,5,9,10,11,12	1,2	The paper work has been completed and it is yet to publish in Annexure 1.
Projects in Specific Application				
10.	Analysis and validation of breaking electric poles during cyclones.	1,2,3,4,5,6,8,9,10,12	1,4	Mr.R.Prabhakaran, Mr.K.Amudhan, Mr.A.Muhamed Ismail Buhari, Mr.V.Karunakaran, Mr.V.Muralidharan, Mr.J.Baskar, Analysis of Breaking Electric Poles during Cyclones, International Web Conference on Smart Engineering Technologies (IWCSET 20), 2020/6/26
11.	Deep learning-based crack detection using Convolutional Neural Network (CNN)	1,2,3,4,5,6,8,9,10,12	1,4	The work will be converted as a conference or Journal paper in the forthcoming academic year.
12.	Increasing Biogas Production in Cow dung by Adding EM Solution.	1,2,3,4,5,6,7,8,9,10,11,12	1,2	The work will be converted as a conference paper in the forthcoming academic year.
13.	Automatic Latching of Door by Sensing Physiognomic Data using Deep Face Recognition Techniques	1,2,3,4,5,6,8,9,10,12	1,4	T. Karthik alias Siva, M S R. Mohan Kumar, D. Manimukilan, K. Hariharan, S. Valai Ganesh, S.Rajakarunakaran, "Automatic Latching of Door by Sensing Physiognomic Data using Deep Face Recognition Techniques", National Conference on Artificial Intelligence Applications (NCAIA'2020), Avinashilingam University, Coimbatore, 2020, Page No.89-93.
14.	Fatigue Strength Analysis and Performance Characteristics of Inconel Material using Wire Arc Additive Manufacturing	1,2,3,4,5,9,10,11,12	1,2	The work will be converted as a conference or Journal paper in the forthcoming academic year.
15.	Performance Analysis and Fabrication of Greenhouse Dryer with Parabolic Trough Collector for Drying Coconuts	1,2,3,4,5,8,9,10,11,12	2	Proposed to modify the design, conduct more experiments and publish in reputed journals.

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16.	Two-wheeler Accident Detection and Emergency Signalling	1,2,3,4,5,8,9,10,11,12	1,3	Phuvenes Kumar P,Venkateshwara Prasad P, Subhash R, International Web Conference on Smart Engineering Technologies 2020 (IWCSET -2020), Ramco Institute of Technology, Rajapalayam. Phuvenes Kumar P- Patent Published in Gazette- Application No:202141001139
Review/Experimental Studies in Emerging Mechanical Sciences				
17.	Strengthening of Argentum by RCS	1,2,3,4,5,9,10,11,12	1,2	P Sureshkumar , C Gururaj, M Harrish, R Kaushik Shrinivas, H Mohammed Safiur Rahman, “Microstructure Studies on Natural state of Argentum by Repetitive Corrugation and Straightening” Journal of Green Engineering (JGE) Volume-10, Issue-7, July 2020Page no. 526–543.
18.	Experimental Study on Mechanical Behaviour of 8 Series Aluminium Metal Matrix Composites.	1,2,3,5,6,8,9,10,11,12	2	The work will be converted as a conference or Journal paper in the forthcoming academic year.
19.	Strengthening of Additive Manufacturing Components by Equal Channel Angular Pressing Process (ECAP)	1,2,3,4,5,9,10,11,12	1,2	The work will be converted as a conference paper in the forthcoming academic year.
20.	Effect of Cryo-machining on precision and machinability of AL6061 alloy	1,2,3,4,5,9,10,11,12	1,2	The work will be converted as a conference or Journal paper in the forthcoming academic year.
21.	Evaluation of structural and electrical properties of synthesized CuO coated transparent FTO using electrolysis.	1,2,4,5,6,7,8,9,10,11,12	2	The work will be converted as a conference or Journal paper in the forthcoming academic year.
22.	Investigation of Mechanical properties of copper and Vanadium Composites	1,2,3,4,6,9,10,12	1,2	Skantha Rajan T., Pooragavan G., Ragulji R., Rakshana Nivaethika, Jabinth J., Selvakumar N., “Investigation of Mechanical Properties of Copper and Vanadium composites”, International Web Conference on Smart Engineering Technologies – 2020, Ramco Institute of Technology, Rajapalayam, 26th & 27th June 2020

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23.	Experimental Study on Mechanical Behaviour of 8 Series Aluminium Metal Matrix Composites	1,2,3,5,6,8,9,10,11,12	2	The work will be converted as a conference or Journal paper in the forthcoming academic year.
24.	Error Analysis of Pick and Place Robotic arm Movement	1,2,3,4,5,9,10,11,12	1,4	The work has been suggested to apply for a patent with added salient features.
25.	Performance Analysis of Flow and Heat Transfer using Pin Fin Dimple Configuration	1,2,3,4,5,6,7,8,9,10,11,12	1,3,4	The work will be converted as a conference paper in the forthcoming academic year.
26.	Analysis of Heat Exchanger with Different Types of Flow paths	1,2,3,4,5,6,7,8,9,10,11,12	1,2,3	Parameswaran M, Sivaraj R, Venkatesh K 2020, Analysis of heat exchanger with different types of flow paths, (IWCSET 2020), Ramco Institute of Technology, Rajapalayam, 27.06.2020.
27.	Experimental Investigation on Surface properties of Yttria Stabilized Zirconia Coated Austenitic Stainless Steel SS316 using Plasma Spray Coating process	1,2,3,4,5,9,10,11,12	1,2	The paper work has been completed and it is yet to publish in Annexure 1.
Solution to Industrial Problems based Projects				
28.	Fire Sniper System for server room in textile industry	1,2,3,4,5,6,8,9,10,11,12	1,2	The industry comes forward to apply their suggestions in future.
29.	Comparison of surface behaviour of HVOF sprayed WC Vs. WC+Cr3C2 composite coating on SS316	1,2,3,4,5,9,10,11,12	1,2	V Kannan, N Logesh Prasanna, C Aravind Kumar, M Ganesh Ram, C Gururaj & S Maharajan, A Review on Thermal Spray Coating Technique on Metal Plates, International Web Conference on Smart Engineering Technologies 2020 (IWCSET -2020), Ramco Institute of Technology, Rajapalayam. Page no. 184-188.
30.	Cycle Time Reduction by Modifying Strainer Bunk	1,2,3,4,5,6,8,9,10,11,12	1,2	The industry expressed interest to implement the modified strainer bunk in their production unit.

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31.	Elimination of Pin Hole side Cast Datum Position over size (pin hole shift) in housing machining process (Calliper)	1,2,3,4,5,6,8,9,10,11,12	2	The solution suggested by the students was appreciated and accepted to implement in Calliper unit.
32.	Automation of Despatching the Cement Packet into the Delivery Area	1,2,3,4,5,6,8,9,10,12	1,4	The industry appreciated the solution given by the students
33.	Experimental Study on Elimination of Tube Gap Defect in Honda Back plate Assembly	1,2,3,4,5,6,8,9,10,12	1,3	The suggestion given by the students to improve the production rate was appreciated and shown interest to implement in Honda Back plate assembly
34.	Improving the Productivity in S-Cam Drum Brake by Positioning the Rivet in Rivering Machine	1,2,3,4,5,6,7,8,9,10,11,12	1,2	This project is implemented in the industry as per the student's suggestion.
35.	Modification of Shear Riverting Machine in Brake Booster to Reduce Stud Thread Shearing	1,2,3,4,5,6,8,9,10,12	1,3	The modification suggested by the students was successfully implemented.